

Object Detection with YOLO

What is Object Detection?

Object detection finds and classifies multiple objects in an image. It outputs bounding boxes (x, y, width, height) and class labels with confidence scores.

YOLO — You Only Look Once

- Single neural network processes entire image at once
- Divides image into $S \times S$ grid cells
- Each cell predicts B bounding boxes + confidence + class probabilities
- Speed: 45 FPS (real-time capable)

Key Metrics

- IoU (Intersection over Union): overlap between predicted and true box
- mAP (mean Average Precision): main evaluation metric
- Confidence Score: $P(\text{object}) \times \text{IoU}$

YOLO Versions

- YOLOv1 (2016): original, 45 FPS
- YOLOv3 (2018): multi-scale detection, much better accuracy
- YOLOv5/v8 (2020-2023): PyTorch, easy to use, state of the art

Quick Code (YOLOv8)

- `from ultralytics import YOLO`
- `model = YOLO('yolov8n.pt')`
- `results = model('image.jpg')`
- `results[0].show()` # display with boxes